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Mechanical properties of Fe–Al–M–C (M = Ti, V, Nb, Ta) alloys with strengthening carbides and Laves phase

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Abstract

The mechanical properties of Fe–15Al (in at.%) and Fe–26Al based alloys with alloying additions are investigated. The alloys contain 1 at.% C and 2 or 4 at.% Ti, V, Nb or Ta. Preliminary information on the mechanical properties of the investigated alloys is provided by means of hardness tests. The mechanical testing of alloys is carried out in compression as well as in tension. Compression tests are carried out at room temperature and at elevated temperatures including the temperature range of the yield stress anomaly. The tensile tests are performed in order to determine the room temperature ductility and high temperature tensile strength at temperatures of possible industrial applications. The investigation of the fracture surfaces of ruptured tensile specimens is performed by means of scanning electron microscopy (SEM).
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1. Introduction

Iron–aluminium alloys are attractive for development of structural materials for use in conventional power plants and in the petrochemical industry, where materials with limited density, reasonable strength and good corrosion resistance are required [1–9]. Low materials costs are an additional advantage for production of Fe–Al alloys. However, there is a need to improve the room temperature ductility and high temperature strength for application of these alloys.

The presence of carbides in Fe–Al alloys was recommended in several studies for a number of reasons. Besides the precipitation strengthening effects, an improvement of the room temperature ductility was observed [10,11]. The studies [12,13] revealed a beneficial effect of carbides on the machinability of iron aluminides. The hard carbide particles in the Fe–Al matrix are also responsible for an improvement of tribological (friction) properties such as wear and erosion resistance [14]. The presence of Laves phase in Fe–Al alloys is recommended due to its very efficient strengthening effect [15–17]. On the other hand, the intermetallic Laves phase precipitates are known to be very brittle [18–20].

The present study concentrates on Fe(Al)-base and Fe_3Al-base alloys with strengthening precipitates, in particular, on the mechanical behaviour of quaternary Fe–15Al–2M–1C and Fe–26Al–(2 or 4)M–1C (M = Ti, V, Nb, Ta) alloys. The binary reference alloys (Fe–15Al and Fe–26Al) are also tested. Whereas the alloys with 15 at.% Al have an A2 bcc disordered matrix up to the melting temperature, the alloys based on Fe–26Al undergo two order–disorder structural transitions, namely $D0_3 \rightarrow B2$ at around 550 °C (depending on alloying additions) and $B2 \rightarrow A2$ at around 800 °C before melting.

Characterisation of constitution and microstructure of most Fe–Al–M–C alloys investigated in this study is reported in [21]. The alloys are usually two-phase, i.e. Fe–Al matrix with MC carbides. An exception is the alloy Fe–26Al–4Nb–1C which was found to be three-phase with the Laves phase $(Fe, Al)_2Nb$ besides NbC carbides in the

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Table 1
Characterisation of microstructure of the investigated alloys

Nominal alloy composition (at.%)	Fe–Al Matrix				Precipitates		
	Grains		Dendrites		Chemical formula	Morphology	Size (μm)
	Morphology	Size (μm)	Morphology	Size (μm)			
Fe–15Al	Polygonal	500	–	–	–	–	–
Fe–15Al–2Ti–1C	–	–	Equiaxed and columnar	30, 30×80	TiC	Fibrous, lamellar (in eutectic)	1.5
Fe–15Al–2V–1C	Columnar	350×2000	–	–	VC	Plate-like (needle-like)	1×8
Fe–15Al–2Nb–1C	–	–	Equiaxed and columnar	30, 30×80	NbC	Granular, lamellar (in eutectic)	<1
Fe–15Al–2Ta–1C	–	–	Equiaxed and columnar	30, 30×80	TaC	Fibrous, lamellar (in eutectic)	<1
Fe–26Al	Polygonal	700	–	–	–	–	–
Fe–26Al–2Ti–1C	Polygonal	120	–	–	TiC	Cube-like	2
Fe–26Al–4Ti–1C	Polygonal	60	–	–	TiC	Cube-like	2
Fe–26Al–2V–1C	Columnar	450×3000	–	–	VC	Plate-like (needle-like)	0.5×15
Fe–26Al–2Nb–1C	Zig–zag	200	–	–	NbC	Cube-like, script-like (in eutectic)	2.5, 1×5
Fe–26Al–4Nb–1C	Polygonal	60	–	–	NbC and $(Fe, Al)_2Nb$	Cube-like, plate-like	1.8, 2×5
Fe–26Al–2Ta–1C	Polygonal and partially zig–zag	200	–	–	TaC	Cube-like, script-like	5, 1

Fe_3Al matrix. Characterisation of all microstructures is summarised in Table 1.

2. Experimental procedure

The alloys were processed by vacuum induction melting. The melts were cast into cylindrical copper moulds (32 mm in diameter by 186 mm in length). For further details the reader is referred to [21].

Brinell hardness was used with a load of 62.5 g and an indentation ball with 2.5 mm diameter. For static mechanical testing in uniaxial compression or tension, the samples were cut from cast ingots by spark erosion. The dimensions of compression samples were $(6\times6\times10)$ mm^3. For tensile tests, flat samples of (3.5×5) mm^2 cross-section with 15 mm gauge length were used. All samples were cut in the direction of the longitudinal axis of the ingot (casting direction). The surfaces of the specimen were ground. All mechanical tests were carried out at a deformation rate of 10^{-4} s^{-1} using a ZWICK 1474 universal testing machine. The mechanical tests at elevated temperatures were carried out in protective argon atmosphere within a radiation furnace. In order to stabilize the heat fluxes and the temperature field, the samples for mechanical tests were held 30 min at the testing temperature before loading. The testing temperature was continuously checked by a NiCr–Ni thermocouple directly on the surface of the test specimen. The fracture surfaces of the tensile samples were investigated by scanning electron microscopy (SEM-Hitachi S-530 with 20 kV, working distance 15 mm) with secondary electrons (SE).

3. Results and discussion

3.1. Mechanical properties at room temperature

Fig. 1 shows the matrix Brinell hardness and the alloy yield stress in compression of the investigated alloys. According to the hardness measurements and compression

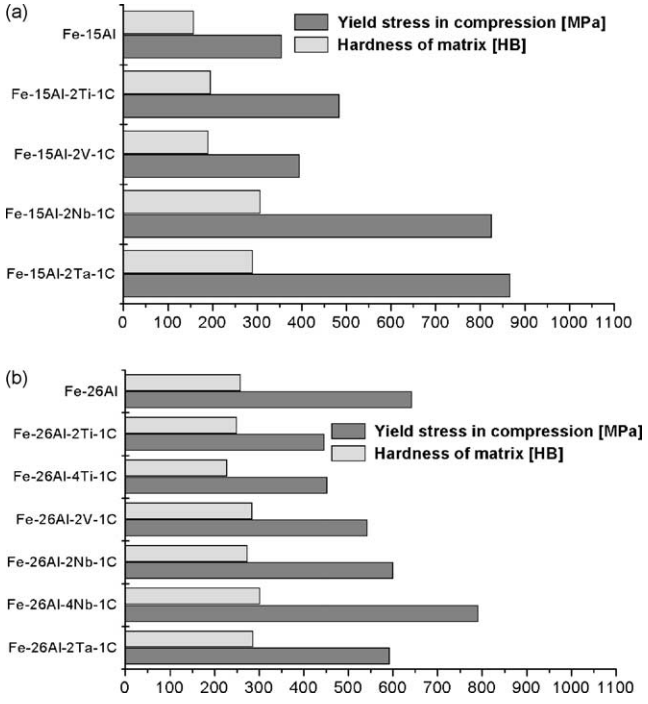


Fig. 1. Yield stress in compression and Brinell hardness of the matrix at room temperature for Fe–15Al-based (a) and Fe–26Al-based (b) alloys.

tests, all Fe–15Al–2M–1C alloys exhibit strengthening of the Fe–Al matrix in comparison to the binary Fe–15Al alloy (Fig. 1a). On the other hand, for Fe–26Al–(2 or 4)M–1C alloys, hardening and also softening effects in comparison to the binary Fe–26Al alloy were revealed by the hardness tests (Fig. 1b). The alloys with V, Nb or Ta show higher hardness than the binary Fe–26Al alloy, whereas the alloys containing Ti are softened. The softening effects of Ti in a single phase Fe_3Al based alloys were also reported in another study [22].

There is a good correlation between hardness and yield stress in compression for Fe–15Al based alloys with Ti and V (Fig. 1a). On the other hand, the alloy Fe–15Al–2Nb–1C has higher hardness, but lower yield stress than the alloy Fe–15Al–2Ta–1C.

The matrix hardness values of most Fe–26Al based alloys do not correlate with values of yield stress in compression (Fig. 1b). Except alloy Fe–26Al–4Nb–1C, all other Fe–26Al based alloys exhibit lower yield stress in compression than the binary Fe–26Al alloy.

Fig. 2 shows the engineering stress–strain curves for the investigated alloys at room temperature in compression. The plastic deformation of Fe–15Al–2M–1C alloys shows parabolic work hardening behaviour (Fig. 2a), which is typical for most disordered polycrystalline metallic materials. The alloys containing Nb or Ta show the highest strengthening effects. The hardening of binary Fe–15Al alloy is low and almost linear.

The deformation behaviour of Fe–26Al–(2 or 4)M–1C alloys (Fig. 2b) exhibits very high strengthening with almost constant hardening rates, until a final stage of parabolic work hardening is reached. The stress–strain curve of the binary Fe–26Al alloy consist of a linear elastic part, followed by a parabolic hardening stage. Despite lower yield stresses of most Fe–26Al based alloys in comparison to the binary alloy, all these alloys exhibit much stronger hardening effects during plastic deformation.

Fig. 3 represents the engineering stress–strain curves of the investigated alloys at room temperature in tension. The ductility of most Fe–Al–M–C alloys is very poor. Some alloys crack even during elastic deformation. Apart from the binary Fe–15Al alloy, the highest values of total elongation were obtained for the alloys Fe–15Al–2Ti–1C and Fe–26Al–2Nb–1C, respectively.

The mechanical properties of investigated alloys tested at room temperature are summarised in Table 2. The tensile tests of both Fe–15Al-based and Fe–26Al-based alloys revealed an earlier onset of plastic deformation than during compression tests, i.e. the values of yield stresses in compression are higher than those in tension. This

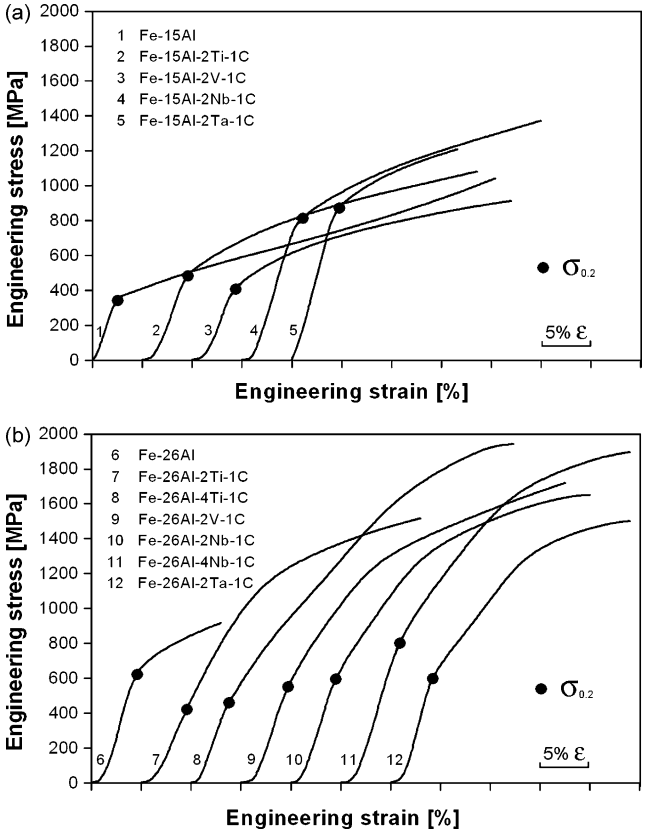


Fig. 2. Engineering stress–strain curves of Fe–15Al-based (a) and Fe–26Al-based (b) alloys tested at room temperature in compression.

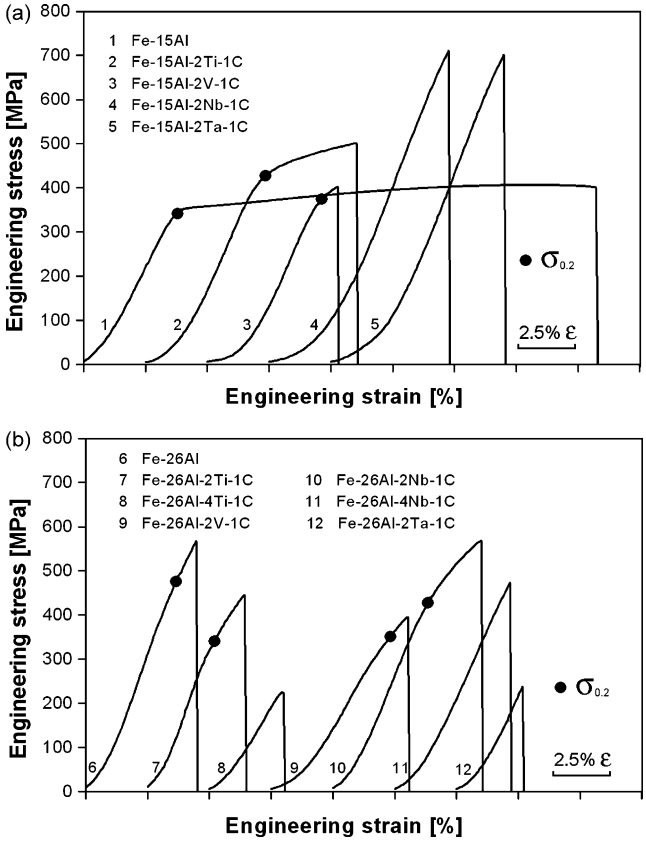


Fig. 3. Engineering stress–strain curves of Fe–15Al-based (a) and Fe–26Al-based (b) alloys tested at room temperature in tension.

Table 2
Mechanical properties and fracture modes of investigated alloys tested at room temperature

Nominal composition (at.%)	Hardness of matrix (HB)	Yield stress in compression (MPa)	Yield stress in tension (MPa)	Total elongation (%)	Mode of fracture
Fe–15Al	159	356	344	16	TGC
Fe–15Al–2Ti–1C	197	485	420	3.3	TGC
Fe–15Al–2V–1C	191	396	374	0.6	TGC
Fe–15Al–2Nb–1C	309	827	–	–	TGC
Fe–15Al–2Ta–1C	291	868	–	–	TGC
Fe–26Al	260	644	475	0.4	TGC
Fe–26Al–2Ti–1C	251	447	342	0.7	IGD+TGC
Fe–26Al–4Ti–1C	229	454	–	–	IGD+TGC
Fe–26Al–2V–1C	286	544	350	0.5	TGC
Fe–26Al–2Nb–1C	275	601	427	1.4	TGC
Fe–26Al–4Nb–1C	303	792	–	–	IGD+TGC
Fe–26Al–2Ta–1C	288	594	–	–	IGD+TGC

TGC, transgranular cleavage; IGD, intergranular decohesion.

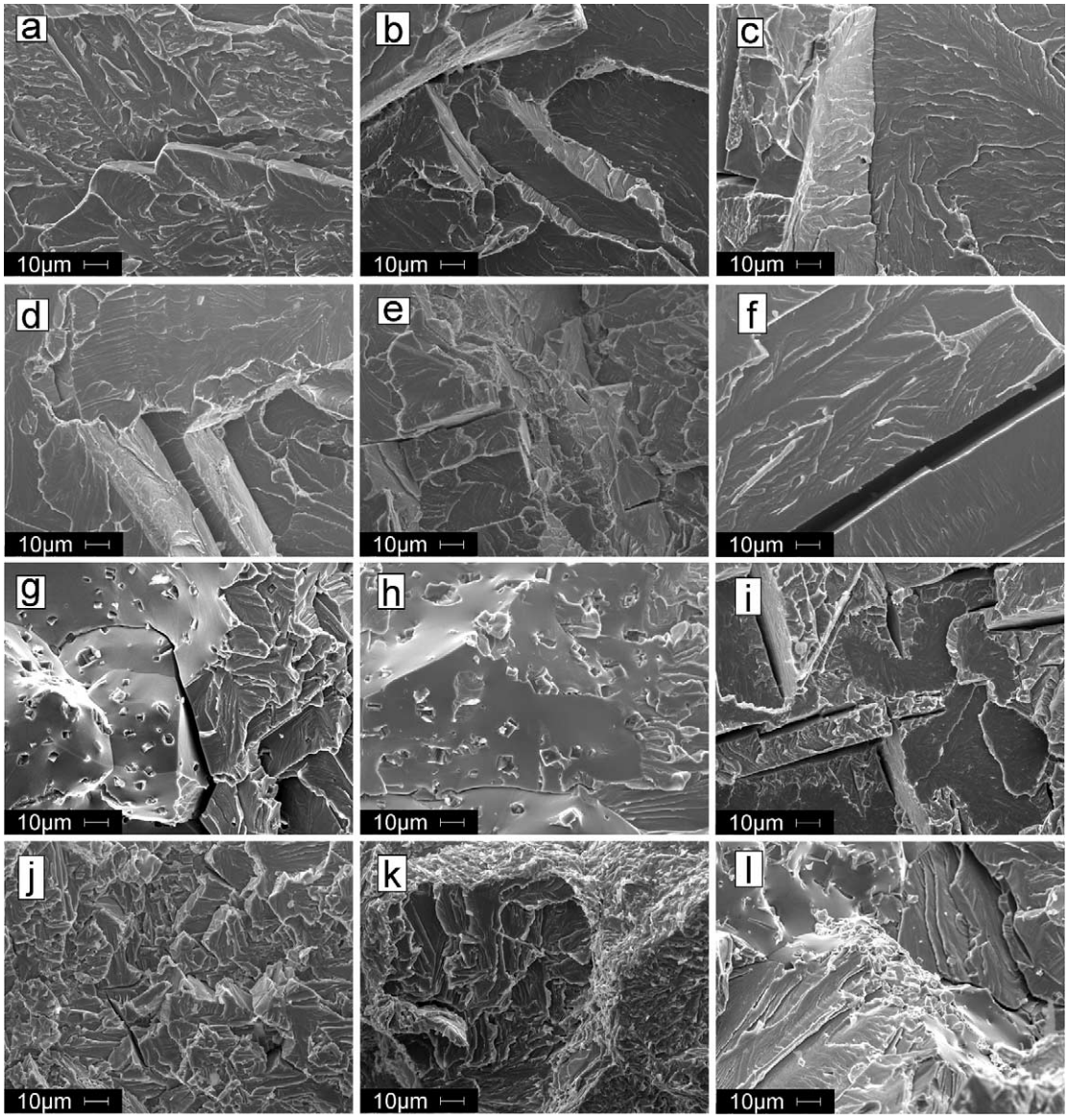


Fig. 4. Fracture surfaces of investigated alloys after tensile testing at room temperature {Fe–15Al (a); Fe–15Al–2M–1C, M=Ti (b), V (c), Nb (d), Ta (e); Fe–26Al (f); Fe–26Al–2M–1C, M=Ti (g), V (i), Nb (j), Ta (l); Fe–26Al–4M–1C, M=Ti (h), Nb (k)}.

phenomenon—the so-called tension–compression asymmetry is known in metals with bcc crystal structure [23–26]. The tension–compression asymmetry has primarily been related to the glide of the screw dislocations with nonplanar (delocalised) core structure [27,28]. Another factor contributing to the tension–compression asymmetry could be the Bauschinger effect, which originates from the different elastic properties of matrix and secondary phases resulting in internal tensions during solidification of alloys. Another significant contribution to the asymmetric tension–compression behaviour is related to the notch sensitivity of the matrix due to the presence of precipitates. In such a case the morphology of the precipitate is the controlling factor for the notching effect in the matrix. Indeed, the alloy Fe–26Al–2V–1C with relatively thin (about 0.5 μm in thickness) and sharp plate-like vanadium carbide precipitates shows a pronounced tension–compression asymmetry (≈36%).

In Fig. 4, SEM-fractographs of investigated alloys are shown. The binary Fe–15Al alloy with ductility of 16%, was fractured with transgranular fracture with very fine cleavage facets (Fig. 4a). All Fe–15Al–2M–1C alloys were fractured with coarse transgranular cleavage (Fig. 4b–e), which corresponds to their brittle behaviour. The alloy Fe–15Al–2Ti–1C exhibits the highest room-temperature ductility (≈3.3%) of all Fe–15Al-based alloys. The binary Fe–26Al alloy exhibits a very coarse transgranular fracture (Fig. 4f). The alloy Fe–26Al–2Nb–1C which has the highest ductility (≈1.4%) of all Fe–26Al based alloys, exhibits a very fine facetted transgranular cleavage (Fig. 4j). The fracture surface of the alloy Fe–26Al–2V–1C with uniform steps indicates transgranular fracture (Fig. 4i), with the influence of the crystallographic orientation of plate-like vanadium carbide precipitates. The orientation relationship of these vanadium carbide precipitates with the matrix was determined in a previous study [21]. According to this result, it can be stated that the fracture facets are formed on the (001) planes of the matrix which are parallel to the (001) planes of the precipitates. For the other Fe–26Al-based alloys a more complex fracture was observed. The fracture surfaces of the alloys with Ti or Ta are partially intergranular and transgranular (Fig. 4g–l). The smooth fracture areas correspond to the decohesion and the facetted areas to the cleavage. On the intergranular fracture areas of the alloys Fe–26Al–2Ti–1C and Fe–26Al–4Ti–1C the cube-like TiC precipitates can be clearly distinguished. The alloy Fe–26Al–4Ti–1C was fractured almost completely with intergranular decohesion. The alloy Fe–26Al–4Nb–1C showed intergranular decohesion besides transgranular cleavage (Fig. 4k). On the intergranular fracture areas many densely distributed precipitates are observed. These precipitates disturbed the intergranular fracture surface so significantly that the decohesion areas are not smooth as usually but rough. The alloys Fe–26Al–4Ti–1C and Fe–26Al–4Nb–1C show totally brittle behaviour at room temperature.

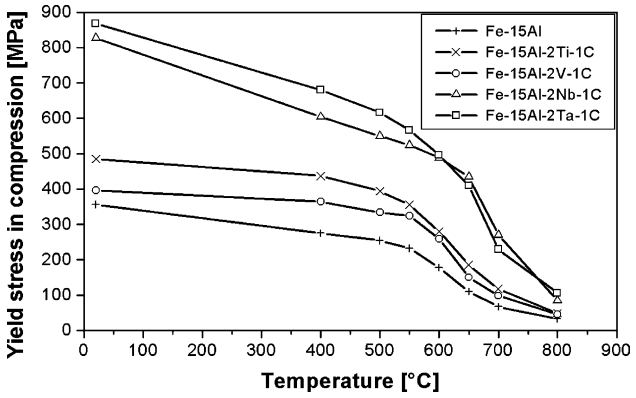


Fig. 5. The temperature dependency of the yield stress in compression for Fe–15Al-based alloys.

3.2. Mechanical properties at elevated temperatures

In order to reveal the potential of investigated alloys as possible high temperature materials, the alloys were tested at elevated temperatures up to 800 °C. Fig. 5 shows the temperature dependency of the yield stress in compression for the Fe–15Al-based alloys. The yield stresses of all Fe–15Al-based alloys show a gradual monotonous decrease with increasing temperature. The alloys with Nb or Ta show the highest strength at all tested temperatures. Fig. 6 shows the temperature dependencies of the yield stress in

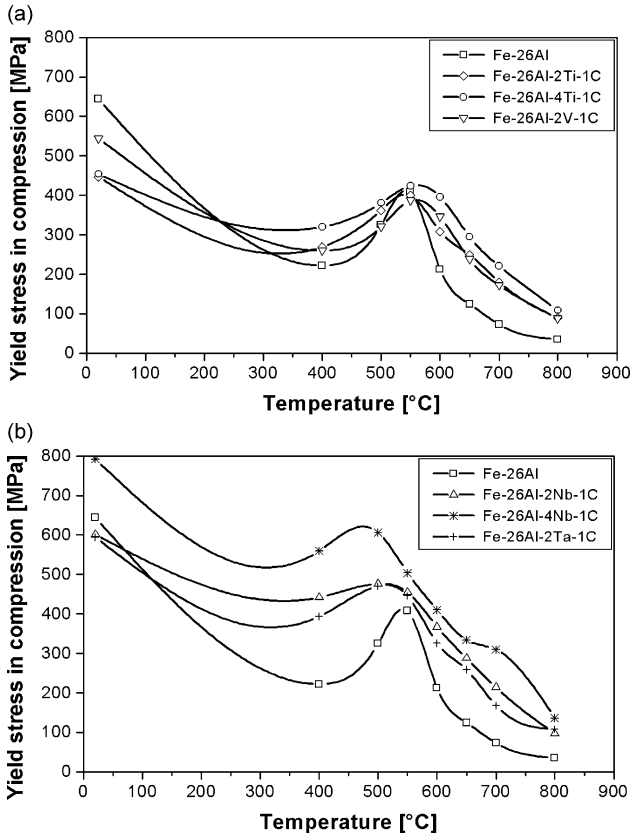


Fig. 6. Temperature dependencies of the yield stress in compression for Fe–26Al-based alloys.

compression for Fe–26Al-based alloys. All Fe–26Al-based alloys exhibit an anomalous increase in yield strength within a certain range of intermediate temperatures. The temperature at which the peak occurs is about 550 °C and the yield stress at the peak maximum is about 400 MPa for the alloys with Ti and V (Fig. 6a). The yield stress anomaly peak of the Fe–26Al-based alloys with Nb or Ta is shifted towards lower temperatures and higher stresses in comparison to the binary Fe–26Al alloy (Fig. 6b). The alloy Fe–26Al–4Nb–1C exhibits the most significant shift of the peak position (474 °C, 619 MPa). This alloy has the highest yield stresses at all tested temperatures, which clearly demonstrates the strengthening effect of the Laves phase (Fe, Al)$_2$Nb. Precipitation strengthening of the Fe–Al matrix through MC carbides was found to be also effective for all alloy compositions under investigation. In addition, strengthening through MC carbides is of high importance because of high temperature stability and higher resistance to coarsening in comparison to the Laves phase.

Since the mechanical testing of alloys at room temperature revealed a significant tension–compression asymmetry, the alloys were tested also in tension at 650 and 800 °C, which are regarded as possible application temperatures. The comparison of the yield stress in compression and in tension at 650 °C (Fig. 7) as well as at 800 °C (Fig. 8) confirms the asymmetric tension–compression behaviour of all investigated alloys at both temperatures.

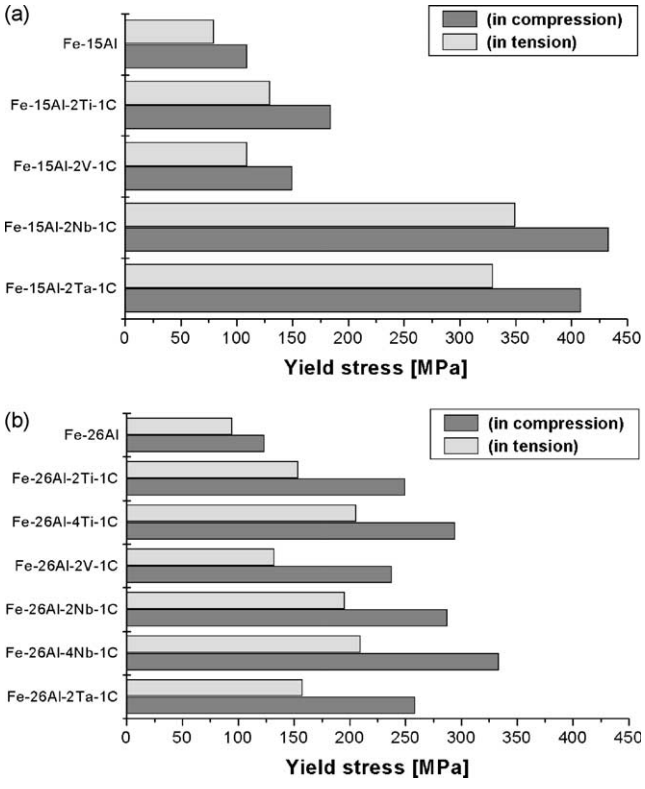


Fig. 7. Yield stresses at 650 °C in compression and in tension for Fe–15Al-based (a) and Fe–26Al-based (b) alloys.

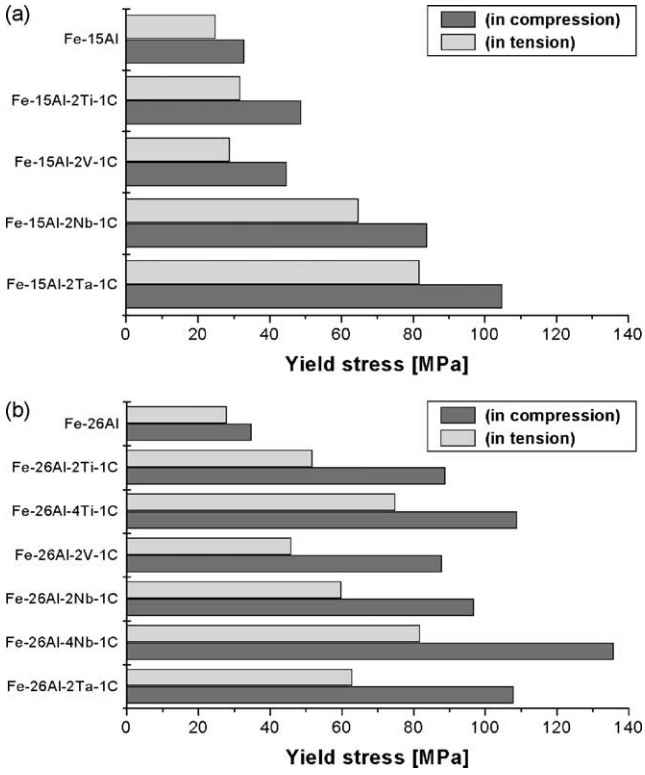


Fig. 8. Yield stresses at 800 °C in compression and in tension for Fe–15Al-based (a) and Fe–26Al-based (b) alloys.

4. Summary and conclusions

Addition of alloying elements such as carbon and transition metals to Fe–Al alloys leads to strengthening or softening effects on hardness and yield stress of the investigated Fe–Al–M–C (M = Ti, V, Nb, Ta) alloys at room temperature. All Fe–15Al-based and Fe–26Al-based alloys exhibit higher hardening rates during plastic deformation than corresponding binary Fe–Al alloys. The alloys with Nb or Ta show the highest strengthening effects. Most investigated Fe–Al–M–C alloys have been found to be rather brittle at room temperature. The fracture surface observations revealed two different fracture modes, namely: transgranular cleavage (TGC) and intergranular decohesion (IGD). Apart from the binary Fe–15Al alloy, the highest values of room temperature ductility were obtained for the alloys Fe–15Al–2Ti–1C (≈3.3%) and Fe–26Al–2Nb–C (≈1.4%).

At elevated temperatures all investigated Fe–Al–M–C alloys exhibit strengthening effects in comparison to the corresponding binary reference alloys. Mechanical testing in compression at various temperatures revealed the usual decrease of the yield stresses with increasing temperature for the Fe–15Al-based alloys. The alloys containing Nb or Ta show the highest strength at all tested temperatures. All Fe–26Al-based alloys exhibit an anomalous increase in strength within a certain range of intermediate

temperatures—the yield stress anomaly. It was found that the alloying elements Ti or V have no significant influence on the position of the yield stress anomaly peak, whereas for the alloys with Nb or Ta the peak is shifted towards lower temperatures. Only the alloy Fe–26Al–4Nb–1C was found to show a substantial strengthening effect on the yield stress at the peak maximum (619 MPa compared to other alloys—ca. 400 MPa). This result demonstrates the strengthening efficiency of Laves phase. All investigated alloys exhibit a tension–compression asymmetry at room temperature as well as at elevated temperatures. The tension–compression asymmetry of Fe–26Al-based alloys is higher than that of Fe–15Al-based alloys. The obtained results show promising properties of the Fe–Al alloys containing Nb or Ta and C for high temperature application.

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